ET TU BRUTE FORCE

Computing power is increasing every day. In fact, it is due to ever increasing computing power that most of the technological advancements of today are possible - IoT, smart homes, autonomous cars and more. With increasing abundant computing power comes the ability to put it to malicious use.

For instance, Okiru is a variant of Mirai that targets ARC processors which are the second most popular processors in the world. Using these botnets that are growing as you read this, breaking into weak password protected accounts is literally not a big deal anymore.

Deployment of AI into brute force methods could change the game altogether. Using generative adversarial networks, where two neural networks are pitted against each other in a match of counterfeiting and detection, it is possible to detect a password much faster than existing methods of brute forcing. Researchers at Stevens Institute of Technology in Hoboken, New Jersey, did exactly that and were able to crack 23 percent of leaked LinkedIn passwords in the first try. Think of this performance in comparison to the recent demonstration of Alpha Go, where the AI came up with strategies that have never been seen before for a board game, and you will get the big picture.

On the other hand, quantum computers are progressing significantly. Quite a lot of leading tech giants today are building and selling (to corporate clients) quantum computers that can deal with double-digit qubit values. While it is true that the rate of progress in quantum computing technology hasn't been remarkable, the same could have been said about the traditional semiconductor processing speeds before the transistor revolution pre-2000. One significant breakthrough could push quantum computers far ahead of their current abilities and that would render every modern day encryption technique quite useless. That's because a quantum computer could run a large number of computations in a small amount of time, making it the perfect tool to brute force your way into a locked system. Even the highly regarded blockchain won't be enough to defy its power.

To be honest, there are ways around this too. For instance, just like quantum computing possesses the power to break down modern day encryption, quantum-encryption techniques will replace them with a far more robust alternative.

Additionally, this same computing power can also be used for other malicious purposes. Brute-force search is something that will progress much faster based on computing advances this year. Due to higher computing power gained via either of the two sources mentioned above, it would become a lot more easier to look anything up online, comparing images to databases of hundreds and thousands and more.

To be appropriately braced against brute force attacks, along with personal security practices, you also need to make informed choices about your cloud service providers and whether or not they're protected against such attacks. Additionally, choose long, random passwords that are harder to brute force. ■

HIT BY THE STUPID STICK

There is a critical vulnerability in humans, called stupidity. One thing that will not change in 2018, is human stupidity. As the social engineering section shows, the individual is the weakest link in the security chain. Being dumb was not much of a problem so far, as everything around us was stupid too. As our television sets, speakers, appliances and dog collars get smart, the way our own stupidity affects us amplifies, and opens up new vectors of attack.

The attacks may not be harmful or devastating, just dumb and hassling. Say a visiting relative plugs in a smart dongle to a stupid television set, and a stupid neighbour watching a stupid WhatsApp forward on his smartphone is stupid enough to cast it by mistake because the option to do so appears on the video playing. Suddenly there is a Malaysian man getting whipped on your television, after which, things will never be the same between you and your visiting relatives.

Cybercriminals taking advantage of stupidity will actually allow them to execute attacks in many of the scenarios listed here. Successful Spear phishing campaigns, database breaches, unsecured cloud storage, bad password practices, social engineering and compromising IoT devices being used with the default settings all depend on the stupidity of the people involved.

One of the commonest kinds of attacks that take advantage of stupidity, is those reports of hacking groups defacing government websites. These kind of attacks typically use some of the oldest tricks in the book, and are executed through simple tools, with little or no coding knowledge required. The defacements primarily depend on finding a website using unpatched technologies, or inadequate security practices. After a particular defacement, it is not likely that the group will ever surface in another incident. These earn no money for the attackers, are trivial to fix, and can be considered as vandalism at most. However, when groups across the border ex-